

Weekly Progress Report

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Domain: Machine Learning / Industrial ML

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I. Overview

This week focused on model refinement, explainability, final evaluation, documentation, and preparation of the project deliverables.

II. Achievements

1. Project 10: Refined the best-performing regression/forecasting approach and evaluated prediction performance across relevant time horizons.
2. Project 10: Investigated feature importance and the effect of excluding highly correlated quality variables where required by the project objective.
3. Project 6: Refined the RUL prediction pipeline and evaluated model performance across the available engine datasets/conditions.
4. Performed error analysis to identify cases where predictions were less reliable and documented possible causes.
5. Prepared plots, tables, methodology notes, and results for the final project report.
6. Organized GitHub repositories with source code, notebooks, README documentation, and report material. **III.**

Challenges

1. Improving accuracy must be balanced against model complexity and interpretability.
2. Generalisation across different operating conditions remains a key challenge in predictive maintenance.
3. Industrial ML results need to be communicated in terms of actionable predictions rather than only model scores.

IV. Learning Resources

1. Project specifications and dataset documentation.
2. scikit-learn and SHAP documentation.
3. Technical references on industrial forecasting, predictive maintenance, and model explainability.

V. Next Week's Goals

1. Complete final documentation and review the repositories.
2. Validate that experiments are reproducible from a clean environment.
3. Prepare final presentation/demo material.
4. Incorporate mentor feedback and document future improvement opportunities.

VI. Additional Comments

The fourth week consolidated the modelling work into reproducible project deliverables and highlighted practical considerations for applying ML to industrial data.